

MR4276287: Uniformity in Mordell-Lang for curves

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Chapter 1

Overview

This chapter gives a concise overview of the geometric and height-theoretic inputs behind the uniform Mordell–Lang bounds for rational points and torsion packets.

Chapter 2

Geometric and height infrastructure

This chapter packages the geometric and height-theoretic input used by the later counting arguments.

2.1 Ambient setup

Definition 1 (Ambient moduli and height data). *Source: ambient setup from the universal-family sections of the paper.* Fix $g \geq 2$. Let $\mathbb{M}_g$ be the fine moduli space of smooth genus- g curves with level structure, let $\mathfrak{C}_g \rightarrow \mathbb{M}_g$ be the universal curve, and let $\mathfrak{A}_g \rightarrow \mathbb{A}_g$ be the universal principally polarized abelian variety with the chosen projective embeddings. Fix the ample line bundles and the associated height functions on the compactifications, together with the fiberwise Néron–Tate height on the universal abelian family.

Lemma 2 (Height comparison on the moduli side). *Source: introduction of the paper, especially the comparison between moduli height and projective height along the Torelli map. The chosen projective height on the moduli space and the height induced by $\mathbb{A}_g$ are comparable up to affine linear error terms along the Torelli map.*

2.2 Betti geometry

Proposition 3 (Betti map for the universal abelian variety). *Source: Betti map construction for the universal abelian variety. Locally on the base, the analytic family of abelian varieties admits a Betti map to $\mathbb{T}^{2g}$ whose restriction to each fiber is a group isomorphism, whose fibers are complex-analytic, and whose product with the projection to the base is a real-analytic isomorphism.*

Proposition 4 (Betti form). *Source: Betti form construction for the universal abelian variety. There is a closed semi-positive $(1,1)$ -form on the analytification of the universal family which scales by N^2 under multiplication-by- N , and whose top wedge detects the maximal real rank of the Betti map on smooth loci.*

Proposition 5 (Betti map without the level-structure restrictions). *Source: appendix extension of the Betti map to arbitrary abelian schemes. The local Betti-map construction persists for arbitrary regular bases and abelian schemes after passing to a finite étale cover when necessary.*

Lemma 6 (Betti rank is insensitive to shrinking the base). *Source: rank stability under Zariski open restriction. Passing to a Zariski open dense subset does not change the maximal Betti-map rank seen on the smooth locus.*

Lemma 7 (Silverman–Tate height comparison). *Source: Silverman–Tate theorem revisited. The canonical height on a fiber differs from the naive projective height by an error bounded linearly in the height of the base point.*

2.3 Height inequality

Definition 8 (Non-degenerate subvariety). *Source: non-degeneracy via Betti rank.* An irreducible subvariety is non-degenerate if some local Betti map has full real rank $2 \dim X$ on a smooth point of the analytification.

Proposition 9 (Auxiliary height inequality). *Source: Siu-style positivity estimate on the graph construction. On a non-degenerate subvariety, multiplication by N gives a lower bound for the projective height of a point in terms of the height of its base point.*

Theorem 10 (Height inequality under the paper’s hypotheses). *Source: proof of the main height inequality under Hyp. After one fixed power of 2 is chosen, the canonical height on the total space is bounded below by a positive multiple of the height on the base on a dense open subset of any non-degenerate subvariety.*

Theorem 11 (Full height inequality). *Source: appendix version without the extra Hyp assumptions. For an arbitrary regular base and a non-degenerate subvariety, the canonical height dominates the height of the image in the base on a dense open subset.*

2.4 Counting lemmas

Lemma 12 (Uniform degree bound). *Source: degree and height control for translated fibers. Translates of a fixed fiber curve have uniformly bounded projective degree, and one translate in each fiber has height controlled by the moduli point.*

Theorem 13 (Non-degeneracy of the Faltings–Zhang image). *Source: Gao’s non-degeneracy theorem for the Faltings–Zhang image. For sufficiently large fiber power, the Faltings–Zhang image of the universal curve family is non-degenerate.*

Lemma 14 (Quantitative Vojta–Mumford estimate). *Source: quantitative Vojta/Mumford bound for large points. A curve inside an abelian variety contributes at most exponentially many points of large canonical height, with the exponent controlled by the subgroup rank.*

Lemma 15 (N-Alon combinatorial lemma). *Source: the combinatorial product-avoidance lemma. Large enough subsets of a curve cannot have all M -fold products trapped in a fixed proper closed subset of the ambient product.*

Lemma 16 (Packets alternative). *Source: the packet-counting lemma via Hurwitz. A proper closed condition on the Faltings–Zhang image can only hold for uniformly many base points on a curve of genus at least 2.*

Proposition 17 (Points on curves are sparse). *Source: Néron–Tate distance proposition for points on curves. For a family of genus- g curves with large moduli height, every point is either exceptional or has only uniformly boundedly many nearby points in the canonical metric.*

Chapter 3

Uniform Mordell–Lang bounds

This chapter derives the uniform bounds on rational points and torsion packets from the geometric and height estimates established earlier.

Proposition 18 (Pre-Mazur bound). *Source: Section 8 of [?]. Above the moduli-height threshold, the non-degeneracy input and the height inequality control the translated curve, and the Vojta–Mumford estimate then bounds the large-height points while the sparse-points proposition handles the remainder.*

Theorem 19 (Uniformity in Mordell–Lang for rational points). *Source: Introduction of [?]. The finite-rank estimate, combined with the height comparison along the Torelli map, gives a uniform bound depending only on the genus and the degree bound.*

Theorem 20 (Finite-rank Mordell–Lang bound). *Source: Introduction of [?]. Once the Jacobian height exceeds the moduli threshold, the pre-Mazur bound applies fiberwise and yields an estimate exponential in the rank.*

Theorem 21 (Uniform torsion packets). *Source: Introduction of [?]. Specializing the finite-rank estimate to the torsion subgroup removes the rank parameter and gives a bound independent of the chosen base point.*